

IEOR 4404 – Fall 2025
Solutions for Practice Midterm (Structural Variant A)

Problem 1 (Even-conditioned Poisson). Let X be $\text{Poisson}(\lambda)$ conditioned on being even, i.e. $\mathbb{P}(X = 2i) \propto e^{-\lambda} \lambda^{2i} / (2i)!$.

(a) Inverse transform via recursion. Write $p_i = \mathbb{P}(X = 2i)$ and $S_{\text{even}} = \sum_{j=0}^{\infty} e^{-\lambda} \lambda^{2j} / (2j)! = e^{-\lambda} \cosh(\lambda)$. Then

$$p_0 = \frac{e^{-\lambda}}{S_{\text{even}}}, \quad \frac{p_{i+1}}{p_i} = \frac{\lambda^2}{(2i+1)(2i+2)}.$$

Algorithm.

1. Generate $U \sim \text{Unif}(0, 1)$. Set $i = 0$, $q = p_0 = \frac{e^{-\lambda}}{S_{\text{even}}}$, and $F = q$.
2. While $U > F$: set $i \leftarrow i + 1$, update $q \leftarrow q \cdot \frac{\lambda^2}{(2i-1)(2i)}$, and $F \leftarrow F + q$.
3. Return $X = 2i$.

This uses $O(\mathbb{E}[X]/2)$ recursion steps on average.

(b) Acceptance–rejection interpretation. Let the proposal be $Y \sim \text{Pois}(\lambda)$ with pmf $g(k) = e^{-\lambda} \lambda^k / k!$ and target

$$f(k) = \begin{cases} \frac{g(k)}{\mathbb{P}(Y \text{ even})}, & k \text{ even}, \\ 0, & k \text{ odd}. \end{cases}$$

Then $\frac{f(k)}{g(k)} = \frac{1}{\mathbb{P}(Y \text{ even})}$ for even k (and 0 otherwise), so $C = \sup_k \frac{f(k)}{g(k)} = \frac{1}{\mathbb{P}(Y \text{ even})}$. The A/R rule “accept iff $U \leq f(Y)/(Cg(Y))$ ” becomes “accept iff Y is even,” and the overall acceptance probability is $1/C = \mathbb{P}(Y \text{ even}) = e^{-\lambda} \cosh(\lambda)$.

Problem 2 (Log-logistic). Let $F(x) = \frac{x^\beta}{1+x^\beta}$ for $x > 0$, $\beta > 0$.

(a) ITM. Set $U = F(X)$. Then $U = \frac{X^\beta}{1+X^\beta}$ so $X^\beta = \frac{U}{1-U}$ and

$$\boxed{X = \left(\frac{U}{1-U} \right)^{1/\beta}}, \quad U \sim \text{Unif}(0, 1).$$

(b) A/R with same family (parameter $\tilde{\beta}$). The densities are $f(x) = \frac{\beta x^{\beta-1}}{(1+x^\beta)^2}$ and $g(x) = \frac{\tilde{\beta} x^{\tilde{\beta}-1}}{(1+x^{\tilde{\beta}})^2}$. The ratio

$$R(x) = \frac{f(x)}{g(x)} = \frac{\beta}{\tilde{\beta}} x^{\beta-\tilde{\beta}} \frac{(1+x^{\tilde{\beta}})^2}{(1+x^\beta)^2}$$

satisfies $R(x) \rightarrow 0$ as $x \downarrow 0$ and as $x \uparrow \infty$ whenever $\tilde{\beta} < \beta$ (heavier tail proposal), hence $\sup_x R(x) < \infty$ and A/R is valid. If instead $\tilde{\beta} > \beta$ (lighter tail proposal), then $R(x) \sim (\beta/\tilde{\beta}) x^{\tilde{\beta}-\beta} \rightarrow \infty$ as $x \rightarrow \infty$, so no finite C exists and A/R fails. For $\tilde{\beta} = \beta$ one can take $C = 1$.

Problem 3 (Uniform in a piecewise region). Let $g_1(x) = 1$ on $0 < x \leq 1$ and $g_2(x) = x^{-2}$ on $x \geq 1$. The total area is $\int_0^1 1 dx + \int_1^\infty x^{-2} dx = 1 + 1 = 2$. Thus the marginal cdf of X for a uniform point (X, Y) in A is

$$F_X(x) = \begin{cases} \frac{1}{2} \int_0^x 1 dt = \frac{x}{2}, & 0 \leq x \leq 1, \\ \frac{1}{2} + \frac{1}{2} \int_1^x t^{-2} dt = 1 - \frac{1}{2x}, & x \geq 1. \end{cases}$$

Inverse cdf:

$$F_X^{-1}(u) = \begin{cases} 2u, & 0 \leq u \leq \frac{1}{2}, \\ \frac{1}{2(1-u)}, & \frac{1}{2} < u < 1. \end{cases}$$

Sampler. Generate $U_1 \sim \text{Unif}(0, 1)$ and set $X = F_X^{-1}(U_1)$. Then generate $U_2 \sim \text{Unif}(0, 1)$ and set $Y = U_2 g(X)$ where $g(X) = 1$ if $X \leq 1$ and $g(X) = X^{-2}$ if $X \geq 1$.

Problem 4 (Composition / mixture). Given

$$f_X(x) = \int_0^\infty \frac{2t}{(1+xt)^3} e^{-t} dt, \quad x \geq 0,$$

view $Z \sim \text{Exp}(1)$ and the conditional density $f_{X|Z=z}(x) = \frac{2z}{(1+xz)^3}$, which is normalized since $\int_0^\infty \frac{2z}{(1+xz)^3} dx = \int_1^\infty \frac{2}{u^3} du = 1$ by $u = 1+xz$. Its cdf is $F_{X|Z=z}(x) = 1 - (1+xz)^{-2}$, hence the inverse

$$\boxed{X = \frac{(1-V)^{-1/2} - 1}{Z}}, \quad Z \sim \text{Exp}(1), V \sim \text{Unif}(0, 1).$$